

Brain storming session (16.04.2026)

will: - AI on the edge ideas - garbage collection - model compression / distillation
- CI/CD (offered to set up the infrastructure for this)

audio visio sound source localization (EFE)

- Will: Imax audio captioning (different whispering models, different languages, different noise levels)
- EFE: put names in front of speaker

Efe interest: physical movement, robotics, agrees for “on the edge” - haukur mentioned physics based: can use CERN data, NASA data

Meteorological data (haukur) - earthquake data - compression of model to run on edge devices (deployment) - interpretability of models (what features are important for the model to make a decision) - confidence of model (how sure is the model about its prediction). Uncertainty of ML - use 1 or 2 sensors, real-time data, online learning, just predict along two dimensions (e.g. temperature and humidity, or pressure and wind speed) - everything is time based, and we can use information from signals & systems

- Efe has problems with earthquakes: what data? what variables

Will: Tribe vs 2 (<https://ai.meta.com/blog/tribe-v2-brain-predictive-foundation-model/>) - take something like this, find a visual medium (e.g. brain rot videos) compress a model to say which region is most likely to activate - compress NLP models to run on edge devices (e.g. mobile phones, IoT devices) - drawing app classifier, make it really well done (haukur: detectify)

EFE: Financial technology - fintech - fraud detection (credit card fraud, insurance fraud) - algorithmic trading (predicting stock prices, optimizing portfolios)

Ideas from will: On edge continual learning - handwriting on tablets: when I get tired my handwriting gets worse - concept: Train an “encoder-decoder” to map noisy handwriting to clean handwriting. But not signal noise, but prototype noise (i.e. bad handwriting to good handwriting). - hopfield networks?

Haukur suggested using gyroscopic information from phones to detect - handwriting - Will: mentioned gait detection (e.g. for stolen phones, or for health monitoring)

Efe’s main point: want the basic idea to be worth 7 points. So its easier to bring it up to a 10 - Novelty, need more novelty in either the problem or the method.

Will: On-the-fly learning w/ spiking neurons? - mentioned refik anadol’s work as inspiration (visualizing gradients and data in real-time, little bit artsy but also very cool)